CYBERSECURITY

First-Semester Practical Syllabus

Foundation → Labs → Projects → Portfolio → Internship

Target learners	B.Tech / undergraduate students beginning cybersecurity
Primary outcome	Become internship-ready through demonstrable practical skills
Primary direction	Red Team / Pentesting, supported by Blue Team, Cloud Security and AppSec
Recommended duration	One semester / approximately 5–6 months

1. Programme Overview

This syllabus is designed to move students from cybersecurity fundamentals to practical, demonstrable capability. The emphasis is not on collecting certificates or completing large numbers of online rooms. Students are expected to learn concepts, reproduce them in controlled environments, experiment, investigate results, document findings, and build a portfolio.

Long-term priority: Red Team / Pentesting > Blue Team / SOC > Cloud Security > AppSec.

Immediate objective: become a technically serious beginner who can understand how security work is performed and can demonstrate that ability through labs and projects.

2. Learning Philosophy

1. Learn – study the concept for 30–60 minutes.
2. Reproduce – perform the same activity yourself.
3. Experiment – change variables and observe the result.
4. Break – test what happens when assumptions fail.
5. Investigate – understand why the system behaved that way.
6. Document – record evidence, reasoning, findings and lessons learned.

All offensive security activities must be performed only against systems the student owns or is explicitly authorized to test.

3. Semester Learning Path

Networking → Linux → Windows → Python → Web Fundamentals → Security Fundamentals → Network Analysis → Basic VAPT → Basic SOC → Projects → Portfolio → Internship Applications

4. Module 1 – Networking Fundamentals

Priority: Highest. Students should understand what happens on a network before attempting security testing.

Topics

- OSI and TCP/IP models
- Ethernet and MAC addresses
- ARP
- IPv4 and IPv6 basics
- Subnetting
- TCP and UDP
- Ports and sockets
- TCP three-way handshake
- DNS and DHCP
- NAT
- ICMP
- Routing and switching
- VLAN fundamentals
- VPN fundamentals
- Firewalls and proxies
- HTTP and HTTPS

Security perspective

- ARP → ARP spoofing concepts
- DNS → DNS attack concepts
- TCP → scanning and connection analysis
- HTTP/HTTPS → web security
- DHCP → network attack concepts
- Firewalls → traffic control and exposure

Exit criteria

Student can explain packet flow, IP addressing, ports, common protocols and basic network security implications without relying on memorized definitions.

5. Module 2 — Linux Fundamentals

- Filesystem and navigation
- Users and groups
- Permissions and ownership
- Processes
- Services
- Networking
- SSH
- Logs
- Package management
- Bash and shell usage
- Cron
- Environment variables
- Basic system administration

Core commands: ls, cd, pwd, cat, less, grep, find, awk, sed, chmod, chown, ps, top, kill, ss, ip, ping, curl, wget, ssh, scp, tar, systemctl, journalctl.
Exit criteria: student can operate and troubleshoot a Linux machine from the command line and investigate basic network/process/log activity.

6. Module 3 — Windows Fundamentals

- Users and groups
- Permissions
- Processes
- Services
- Windows Firewall
- Event Viewer
- PowerShell
- Registry basics
- Windows networking
- Authentication
- NTFS
- Security logs

Active Directory Fundamentals

- Domain and Domain Controller
- Users and groups
- Organizational Units
- Group Policy
- Kerberos
- LDAP
- NTLM
- Network shares
- Domain authentication

Advanced Active Directory exploitation is outside the initial foundation phase. The priority is understanding how the environment works.

7. Module 4 – Python for Security Automation

- Variables, conditions and loops
- Functions
- Lists and dictionaries
- File handling
- Exceptions
- Modules
- HTTP requests
- JSON
- Regular expressions
- Sockets
- Command-line arguments
- Logging and basic error handling

Expected outcome: write small security utilities independently rather than copying existing tools.

8. Module 5 – Web Fundamentals

Students should understand the complete request path:

Browser → HTTP Request → Web Server → Application → Database

- HTTP methods
- Headers and status codes
- Cookies and sessions
- Authentication and authorization
- Parameters and forms
- REST APIs
- JSON
- Basic JavaScript
- SQL basics

Security topics

- SQL Injection
- Cross-Site Scripting (XSS)
- Broken authentication
- Broken authorization
- Path traversal
- Insecure file upload

9. Module 6 – Security Fundamentals

- CIA triad
- Authentication, authorization and accounting
- Least privilege
- Defense in depth
- Vulnerability, threat, risk and exploit
- Attack surface
- Security controls
- Encryption
- Hashing
- Digital signatures
- Certificates and PKI
- MFA
- Zero Trust fundamentals

Students should be able to explain why a vulnerability exists, not merely name it.

10. Module 7 – Core Security Tools

Tool	Core skills	Expected outcome
Wireshark	Packet capture, filtering, protocol analysis	Explain what happened on the network
Nmap	Host discovery, port scanning, service/version detection, basic scripting	Perform authorized reconnaissance
Burp Suite	Proxy, HTTP history, Repeater, Intruder basics, request/response analysis	Inspect and modify web traffic
Linux CLI	tcpdump, curl, ss, netcat, grep, awk, sed	Investigate and troubleshoot systems
SIEM	Log collection, search, detection, alert and investigation	Understand basic SOC workflow

11. Practical Projects

Project 1 — Home Network Security Lab

Duration: Month 1

Build a virtual network with Linux, Windows and a security VM. Practice IP addressing, subnetting, ports, routing, DNS and firewall rules. Capture DNS, TCP, HTTP/HTTPS and ICMP traffic with Wireshark.

Deliverable: Network diagram; IP table; packet captures; security observations; hardening recommendations.

Project 2 — Python Network Recon Tool

Duration: Month 2

Build an authorized reconnaissance utility: host availability → port scan → basic service identification → report. Add CLI arguments, JSON output, logging, timeouts and basic banner collection as appropriate.

Deliverable: GitHub repository; README; source code; usage guide; screenshots; sample output; limitations.

Project 3 — Web Security Lab

Duration: Month 3

Deploy or build a deliberately vulnerable web application in a private lab. Study controlled examples of SQLi, XSS, broken authentication/authorization, path traversal and insecure file upload.

Deliverable: For each finding: vulnerability, cause, reproduction, impact, remediation and defensive detection.

Project 4 — SOC / Log Analysis Lab

Duration: Month 4

Collect Linux/Windows activity in a lab SIEM. Investigate failed/successful logins, user creation, suspicious processes, network connections and file activity.

Deliverable: Incident reports containing timeline, source, affected host, user, evidence, analysis, severity, containment and prevention.

Project 5 — Windows + Active Directory Lab

Duration: Months 5–6

Build a small domain environment with a Domain Controller and clients. Study authentication, permissions, GPO, Kerberos, LDAP, NTLM, shares and Windows event logs; later introduce controlled attack-and-detect scenarios.

Deliverable: Architecture; lab setup; attack/detection notes; evidence; defensive recommendations.

Project 6 – Mini Cloud Security Lab
Duration: Later / optional in semester

Deploy a small AWS/Azure environment after the foundation is strong. Study IAM, roles, permissions, network configuration, security groups, logging and storage security.

Deliverable: Document safe misconfigurations, detection methods and remediation.

12. Project Quality Standard

A project is not complete merely because it works. Each serious project should follow:

Problem → Architecture → Implementation → Testing → Security Testing → Findings → Fixes → Documentation

- README: what the project is and why it exists
- Architecture diagram
- Technical explanation and design decisions
- Screenshots/evidence
- Security analysis
- Lessons learned
- Future improvements

13. Hands-on Learning Platforms

- TryHackMe
- Hack The Box
- PortSwigger Web Security Academy
- OverTheWire
- picoCTF

Platforms are laboratories, not substitutes for understanding. Depth of learning matters more than the number of rooms completed.

14. Semester Schedule

Period	Primary focus	Project	Career activity
Month 1	Networking + Linux	Home Network Security Lab	Create GitHub structure
Month 2	Windows + Python	Python Recon Tool	Begin internship research
Month 3	Web + Security Fundamentals	Web Security Lab	Portfolio documentation
Month 4	Network analysis + SOC	SOC / Log Analysis Lab	Begin applications if criteria are met
Months 5–6	Active Directory + deeper VAPT/SOC	AD Security Lab	Interview preparation + continued applications

15. Recommended Weekly Allocation

Area	Approx. time
Fundamentals	40%
Hands-on labs	25%
Projects	25%
Documentation / career	10%

16. Portfolio & GitHub Requirements

- cybersecurity-home-lab
- python-network-recon
- web-security-lab
- soc-log-analysis
- active-directory-security-lab
- cloud-security-lab

Do not publish credentials, secrets, private data, employer information, or security material that you are not authorized to disclose.

17. Internship Preparation

Target roles, in priority order:

7. VAPT / Pentesting Intern
8. Cybersecurity Intern – VAPT + SOC
9. SOC Intern
10. Security Analyst Intern
11. Network Security Intern
12. Security Research Intern
13. AppSec Intern
14. Cloud Security Intern

Begin applications once the student can demonstrate networking/Linux fundamentals, 2–3 completed projects, several hands-on labs, and basic Nmap, Wireshark, Burp Suite and Python skills. Learning continues during the application process.

18. Certification Guidance

- CCNA – recommended after genuine networking practice
- Security+ – after networking, Linux, Windows and security foundations
- eJPT → PNPT / OSCP – possible later offensive-security progression
- CySA+ – possible later blue-team/SOC specialization

Certifications support practical ability; they do not replace it.

19. First-Semester Exit Criteria

Area	Exit requirement
Networking	Explain what happens when a URL is entered into a browser; understand IPs, ports, protocols and packet flow.
Linux	Operate and troubleshoot a Linux machine from the CLI.
Windows	Understand basic processes, services, permissions and logs.
Python	Write a small security automation script independently.
Web	Explain browser → server → backend → database communication.
Red Team	Perform basic authorized reconnaissance and vulnerability analysis in a lab.
Blue Team	Read logs and investigate suspicious activity.
Security	Explain why a vulnerability exists and how it can be fixed/detected.
Professional	Explain a project from architecture → implementation → security → findings → fixes.

20. Milestone 1 – Internship Readiness

- Strong networking fundamentals
- Comfortable Linux usage
- Basic Windows administration
- Basic Python
- Web fundamentals
- Security fundamentals
- Working knowledge of Wireshark, Nmap and Burp Suite
- Basic SOC concepts
- 3+ documented projects
- Active GitHub portfolio
- Hands-on lab experience
- Resume ready
- Internship applications underway

The milestone is achieved when students can demonstrate knowledge instead of merely claiming it.

21. Long-Term Progression

1st internship → basic cybersecurity exposure → VAPT/SOC experience → more technical internship → enterprise security exposure → advanced security internship → deeper specialization.

Progress should be judged primarily by technical difficulty, responsibility and mentorship, with company reputation as an additional factor.

22. North Star

The goal of the first semester is not to produce an advanced pentester. It is to produce a technically competent beginner whom a cybersecurity team can take a chance on as an intern.

Learn → Build → Break → Investigate → Document → Apply → Internship → Learn from professionals → Upgrade skills → Better opportunity → Deeper specialization

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The guiding question throughout the programme is: "What skill would make me more capable than I was yesterday, and what can I build to prove I learned it?"